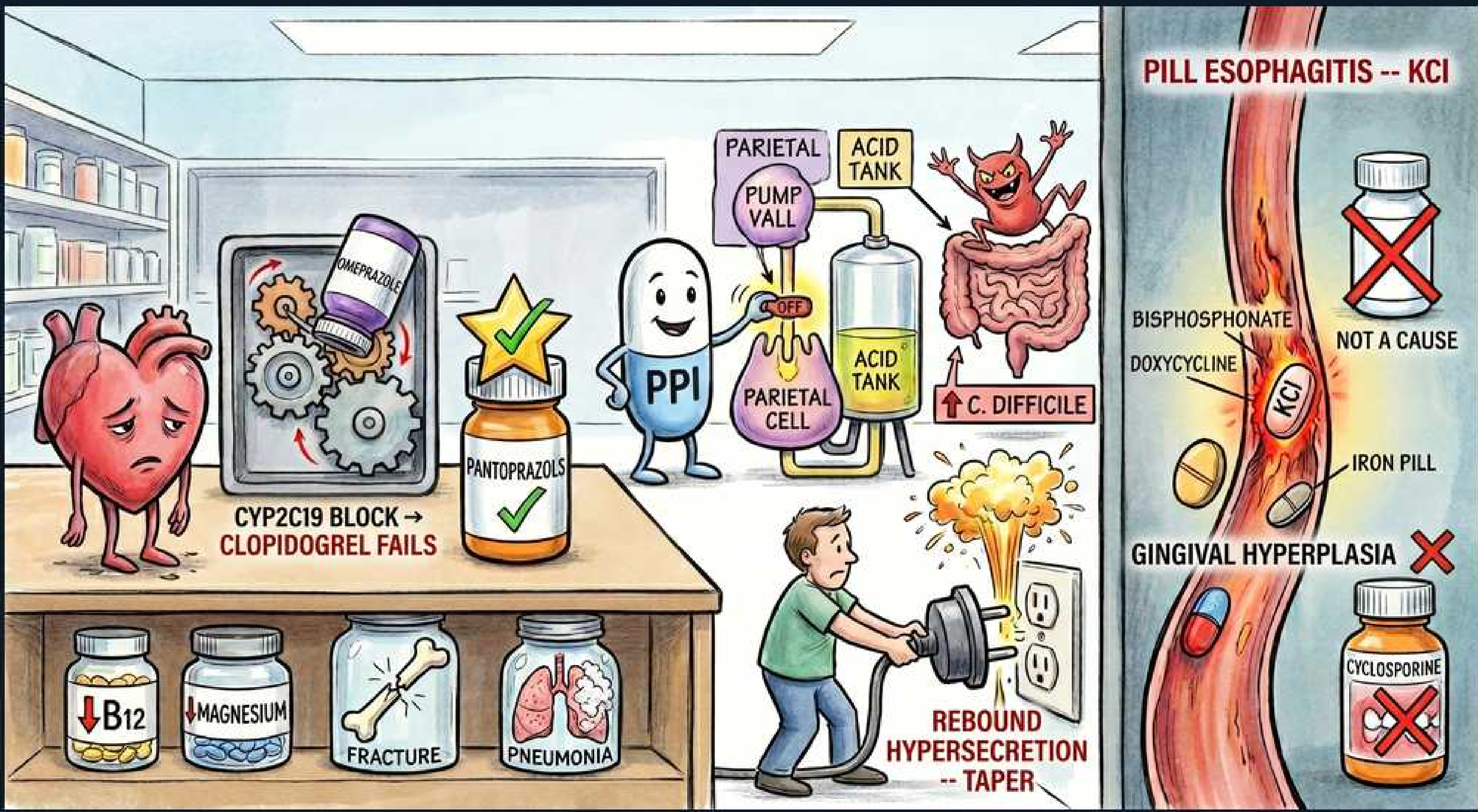
PUD — PPI Pharmacology, Harms & Drug-Induced Injury



1. PPI mascot shutting the proton pump

WHAT YOU SEE
A friendly PPI capsule turning off a proton-pump valve on a parietal cell; the acid tank drops.

WHAT IT ENCODES
PPIs irreversibly inhibit the H+/K+ ATPase (the final common step of acid secretion) on the parietal cell → most potent acid suppression available. They are prodrugs activated in acid, so they work best taken ~30 min before meals.

BOARD TRAP
'Take the PPI after a meal / at bedtime on an empty stomach for best effect' → wrong; give ~30 min BEFORE a meal so active pumps are inhibited.

2. Clopidogrel soldier weakened + CYP2C19 gears jammed

WHAT YOU SEE
A deflated heart-shaped clopidogrel soldier; CYP2C19 gears jammed by an omeprazole bottle — CYP2C19 BLOCK → CLOPIDOGREL FAILS.

WHAT IT ENCODES
Clopidogrel is itself a prodrug needing CYP2C19 to become active. Omeprazole and esomeprazole strongly inhibit CYP2C19 → reduced antiplatelet effect, a classic interaction.

BOARD TRAP
'All PPIs interact equally with clopidogrel' → wrong; omeprazole/esomeprazole are the strong CYP2C19 inhibitors — pantoprazole is preferred.

3. Pantoprazole (gold star, green check)

WHAT YOU SEE
A pantoprazole bottle with a gold star and green check — SAFEST WITH CLOPIDOGREL. (Label misspelled 'pantoprazols'.)

WHAT IT ENCODES
Pantoprazole has the least CYP2C19 inhibition, so it's the PPI of choice when a patient also needs clopidogrel.

BOARD TRAP
'Avoid PPIs entirely in any clopidogrel patient' → wrong; if a PPI is indicated, choose pantoprazole rather than dropping acid protection.

4. C. difficile imp

WHAT YOU SEE
A grinning C. difficile imp climbing out of the gut as the acid barrier drops — ↑ C. DIFFICILE.

WHAT IT ENCODES
Loss of the gastric acid barrier on PPIs increases risk of enteric infections, notably C. difficile and other gut pathogens.

BOARD TRAP
'PPIs lower C. difficile risk by sterilising the gut' → wrong; they RAISE the risk by removing the acid barrier.

5. Rebound acid hypersecretion

WHAT YOU SEE
A patient yanking the PPI plug from the wall; an acid geyser erupts — REBOUND HYPERSECRETION — TAPER.

WHAT IT ENCODES
Chronic PPI use raises gastrin → parietal-cell/ECL hyperplasia. Stopping abruptly causes rebound acid hypersecretion and symptom recurrence → taper rather than stop suddenly.

BOARD TRAP
'PPIs can always be stopped abruptly with no consequence' → wrong; abrupt withdrawal causes rebound hyperacidity — taper.

6. Long-term harm jars

WHAT YOU SEE
Shelf of jars: ↓B12, ↓MAGNESIUM, FRACTURE, PNEUMONIA.

WHAT IT ENCODES
Long-term PPI associations to know: reduced B12, iron and calcium absorption (→ fracture risk), hypomagnesaemia, and increased community-acquired pneumonia; possible acute interstitial nephritis. Acid is needed to release B12 and absorb non-haem iron/calcium.

BOARD TRAP
'Long-term PPIs improve iron and B12 absorption' → wrong; they REDUCE absorption of B12, iron and calcium.

7. Pill esophagitis — KCl

WHAT YOU SEE
A raw, ulcerated oesophagus with a KCl pill stuck halfway down — PILL ESOPHAGITIS — KCl; bisphosphonate, doxycycline and iron pills also lodged.

WHAT IT ENCODES
Caustic pills that lodge in the oesophagus cause focal ulceration: classically potassium chloride, bisphosphonates, doxycycline/tetracyclines, NSAIDs and iron. Prevent it by taking pills with plenty of water and staying upright.

BOARD TRAP
'Pill oesophagitis is from acid reflux, treat with a PPI' → wrong; it's direct caustic mucosal injury — take pills upright with water; the offending drug is the cause.

8. Red X on acetaminophen — NOT a cause

WHAT YOU SEE
An acetaminophen bottle with a big red X — NOT A CAUSE.

WHAT IT ENCODES
Paracetamol/acetaminophen is GI-sparing: it does not inhibit COX in the gut mucosa, so it does not cause peptic ulcers or pill oesophagitis. It's the safe analgesic choice in ulcer-prone patients.

BOARD TRAP
'Acetaminophen causes peptic ulcers like NSAIDs' → wrong; it's GI-sparing and the preferred analgesic in ulcer risk.

9. Cyclosporine → gingival hyperplasia (red X)

WHAT YOU SEE

A cyclosporine bottle pointing to swollen gums labelled GINGIVAL HYPERPLASIA, with a red X.

WHAT IT ENCODES

Cyclosporine causes gingival hyperplasia (also seen with phenytoin and calcium-channel blockers) — included here as a distractor: it is NOT a cause of peptic ulcer or pill oesophagitis.

BOARD TRAP

'Cyclosporine is a cause of drug-induced peptic ulcer' → wrong; its classic adverse effect is gingival hyperplasia, not ulcers.